Student Engagement & Achievement Analysis

A Project Report submitted in partial fulfillment of the

requirements for the award of degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted

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ADITYA UNIVERSITY

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ADITYA UNIVERSITY

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CERTIFICATE

This is to certify that the project work entitled *"**Student Engagement & Achievement Analysis**"* is being submitted by

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in partial fulfillment of the requirements for award of the B.Tech. degree in Artificial Intelligence and Machine Learning.

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DECLARATION

We hereby declare that the project entitled **"Student Engagement & Achievement Analysis"** is a genuine project. This work has been submitted to the **ADITYA UNIVERSITY (Formerly Aditya Engineering College (A)),** Surampalem in partial fulfillment of the **B.Tech.** degree**.** We further declare that this project work has not been submitted in full or part of the award of any degree or any other educational institutions.

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ABSTRACT

The modern education system is evolving beyond academic performance, where student success now depends on both academics and co-curricular engagement. However, most institutions only track participation as attendance, not the quality or impact of involvement, leading to an "Engagement-Achievement Gap." This gap makes it difficult to understand which activities truly contribute to skill development and recognition. This project aims to bridge that gap using data mining techniques by analyzing the relationship between student engagement and achievement. It evaluates factors such as event category, role, and consistency of participation to identify meaningful patterns. By moving from traditional reporting to predictive analysis, the system uncovers key behaviors that influence student success. Machine learning techniques are used to gain deeper insights, where clustering methods like K-Means group students into different engagement profiles, and classification methods like Decision Trees predict achievement outcomes. The model helps distinguish high achievers from passive participants and identifies important factors such as leadership roles and active participation. It also highlights high-impact events that contribute significantly to student growth. This enables institutions to improve planning, optimize resource allocation, and replace low-impact activities with more effective ones. The system acts as a decision-support tool for administrators, providing data-driven insights for enhancing student engagement strategies. At the same time, students benefit by understanding how to improve their participation and build stronger profiles for future careers. Overall, the project promotes data-driven decision-making in education and ensures that co-curricular activities become a key factor in achieving student success.

Keywords:

- Student Engagement
- Data Mining
- Machine Learning
- Achievement Prediction

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1. INTRODUCTION

1.1 Introduction

Student engagement plays a vital role in determining academic success. This project uses machine learning techniques to analyze student engagement data and predict achievement levels. WEKA is used as the primary tool for preprocessing and classification. The system helps in identifying patterns and relationships between engagement factors and performance.

1.2 Problem Statement

In traditional systems, student performance analysis is done manually, which is time-consuming and prone to errors. It is difficult to identify key factors affecting student achievement using manual methods. Therefore, there is a need for an automated system that can efficiently analyze data and provide accurate predictions.

1.3 Objectives

- To analyze student engagement data
- To preprocess the dataset using WEKA
- To convert categorical data into machine-readable format
- To build a classification model using J48 algorithm
- To predict student achievement levels

1.4 Scope

The system can be used in educational institutions to analyze student behavior and performance. It helps teachers and administrators understand how engagement impacts achievement and supports better decision-making.

2. SYSTEM OVERVIEW

2.1 Existing System

Currently, student data is analyzed manually or using basic tools like spreadsheets. There is no intelligent system to predict student performance based on engagement factors.

2.2 Limitations of Existing System

- Time-consuming process
- Prone to human errors
- Difficult to analyze large datasets
- No predictive capability

2.3 Proposed System

The proposed system uses WEKA to preprocess the dataset and apply machine learning algorithms. The J48 decision tree classifier is used to predict student achievement based on engagement attributes. This system automates data analysis and provides better insights.

2.4 Software Requirements

The system requires WEKA tool for data mining, along with Java environment. CSV/ARFF files are used for dataset handling. Basic operating system like Windows is sufficient.

2.5 Hardware Requirements

A basic computer system with at least 4GB RAM and standard processor is sufficient. No high-end hardware is required. It can run on any normal desktop or laptop.

3. TECHNOLOGIES USED

3.1 WEKA

WEKA is a data mining tool used in this project to perform data preprocessing and classification. It provides various filters such as ReplaceMissingValues, NominalToBinary, and Normalize to clean and transform the dataset. The J48 decision tree algorithm available in WEKA is used to build a classification model for predicting student achievement.

3.2 Java

Java is required to run WEKA since it is developed using the Java programming language. It provides platform independence and supports the execution of WEKA tools efficiently. Java ensures smooth functioning of the system environment used in this project.

3.3 CSV Files

CSV (Comma Separated Values) files are used to store and manage the student engagement dataset in a tabular format. These files are easy to handle and can be directly imported into WEKA for preprocessing and analysis.

3.4 ARFF Format

ARFF (Attribute Relation File Format) is the standard file format used by WEKA. It stores both data and metadata, making it suitable for machine learning tasks. The dataset is converted into ARFF format before applying algorithms.

3.5 ETL Process

ETL (Extract, Transform, Load) process is used to prepare the dataset for analysis. It involves extracting raw student engagement data, cleaning it by handling missing values, and transforming it into a structured format suitable for machine learning.

4. DATABASE DESIGN

4.1 Database Table

Student Engagement Table

This table stores all student engagement attributes along with the predicted achievement level.

4.2 Table Structure

Column Name	Data Type	Description
StudentID	VARCHAR	Unique identifier for each student
Dept	VARCHAR	Department of the student
Category	VARCHAR	Type of activity (Technical, Cultural, etc.)
Scale	VARCHAR	Level of activity (Department, College, National)
Participation	VARCHAR	Type of participation (Volunteer, Organizer, etc.)
Freq	INT	Frequency of participation
Time_Inv	FLOAT	Time invested in activities
Prior_Exp	VARCHAR	Prior experience (Yes/No)
Motivation	VARCHAR	Reason for participation (Skill, Social, etc.)
Role	VARCHAR	Role type (Individual/Team)
Mentor	VARCHAR	Mentorship support (Yes/No)
Achievement	VARCHAR	Target variable (High Achiever, Participant, etc.)

Table 4.1: Student Engagement Table

4.3 Keys and Constraints

4.3.1 Primary Key

- StudentID → Unique identifier for each student record

4.3.2 Constraints

- **NOT NULL** → Dept, Category, Participation, Motivation, Achievement
- **CHECK Constraints:**
- Prior_Exp IN ('Yes', 'No')
- Role IN ('Individual', 'Team')
- **Range Constraints (logical):**
- Freq ≥ 0
- Time_Inv ≥ 0
- **StudentID should be UNIQUE**

4.4 Table relationships:

No relationships (Single table only)

Since all data is in one dataset, no need for multiple tables

This is a flat table structure, suitable for data mining using WEKA

4.5 Sample Data / Sample Record:

Dept	Category	Participation	Freq	Time_Inv	Motivation	Role	Achievement
CSE	Technical	Volunteer	3	5.5	Skill Dev	Individual	High Achiever
ECE	Cultural	Attendee	1	2.0	Social	Team	Participant
CSE	Sports	Organizer	4	6.5	Leadership	Team	Consistent Contributor
MECH	Technical	Volunteer	2	4.0	Skill Dev	Individual	Participant
EEE	Cultural	Attendee	1	1.5	Social	Team	Participant

Table 4.2: Sample Student Engagement Data

5. SYSTEM MODULES

Module 1: Data Collection

This module is responsible for collecting student engagement data such as department, category of activity, participation level, motivation, role, prior experience, and time investment. The dataset is stored in CSV format and used for further processing in WEKA.

Module 2: Data Preprocessing (ETL)

In this module, the raw dataset is cleaned and prepared for analysis. Missing values are handled, irrelevant attributes such as StudentID are removed, and categorical data is transformed into a structured format suitable for machine learning using WEKA filters.

Module 3: Data Transformation

The cleaned dataset is converted into ARFF format and transformed using techniques such as NominalToBinary conversion and normalization. The target attribute (Achievement) is properly defined for classification.

Module 4: Model Building (J48 Algorithm)

This module applies the J48 decision tree algorithm to the processed dataset. The model is trained using student engagement attributes to classify and predict achievement levels. The decision tree structure helps in understanding relationships between attributes.

Module 5: Model Evaluation

The performance of the classification model is evaluated using 10-fold cross-validation. Metrics such as accuracy, precision, recall, and confusion matrix are used to assess the prediction performance.

Module 6: Result Visualization

The final decision tree is visualized using WEKA. The visualization clearly shows how different engagement attributes such as participation, motivation, and role influence student achievement.

6. IMPLEMENTATION

6.1 Input Validation

Before processing the dataset in WEKA, the input data is validated to ensure correctness and consistency. Missing values are handled, and invalid or inconsistent entries are removed. Categorical attributes such as participation, role, and prior experience are checked to ensure they contain valid values. This step ensures that only clean and reliable data is used for analysis.

6.2 Exception Handling

During preprocessing and model building, WEKA handles exceptions such as missing values, incompatible data types, and invalid attribute formats. These issues are resolved using preprocessing filters like ReplaceMissingValues and NominalToBinary. This ensures smooth execution of the data mining process without errors.

6.3 Data Processing and Transformation

The dataset is processed using WEKA filters to prepare it for machine learning. Irrelevant attributes such as StudentID are removed. Categorical data is converted into binary format using NominalToBinary, and normalization is applied to scale numeric values. The dataset is then converted into ARFF format for compatibility with WEKA.

6.4 Model Implementation

The J48 decision tree algorithm is applied to the preprocessed dataset. The model is trained using student engagement attributes, and 10-fold cross-validation is used for evaluation. The decision tree is generated to classify students into different achievement levels.

6.5 Result Generation

The output of the model includes classification accuracy, confusion matrix, and decision tree visualization. These results help in understanding the performance of the model and the influence of different attributes on student achievement.

7. OUTPUT SCREENS

The following section presents the output screens of the system. These screenshots demonstrate the preprocessing steps, classifier selection, and results obtained using WEKA. Each screen represents a stage in the implementation of the project.

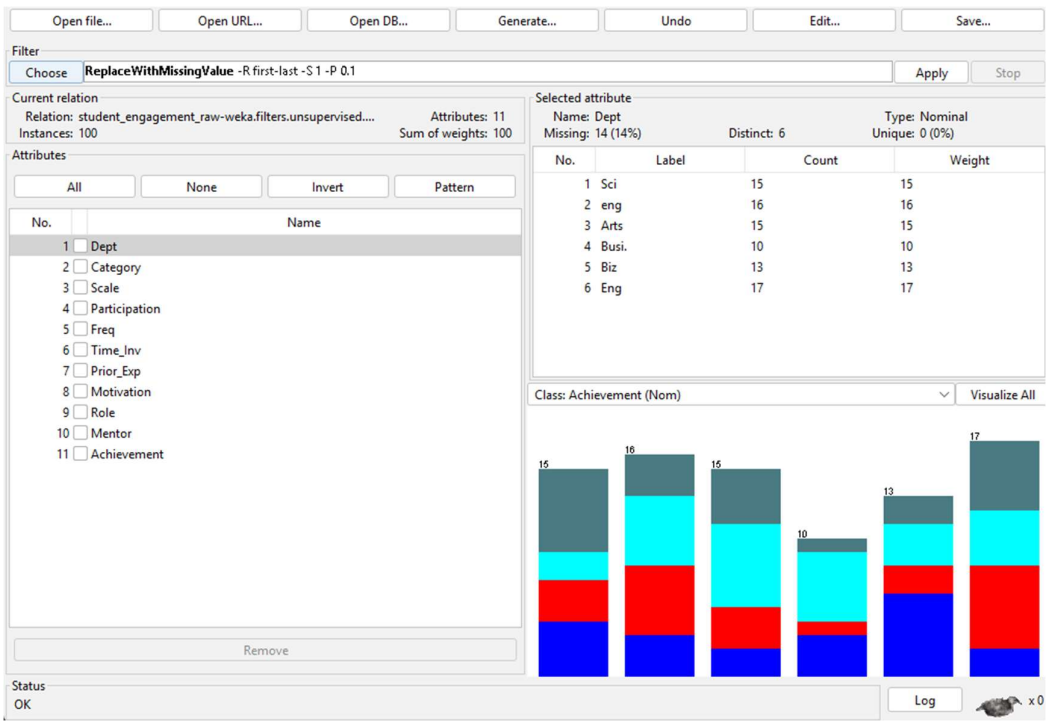

Figure 7.1: *Handling Missing Values using ReplaceMissingValues Filter in WEKA*

The dataset is loaded into WEKA's Preprocess tab, where different student engagement attributes are displayed. The **ReplaceMissingValues filter** is selected to handle missing data, ensuring data completeness. Statistical details such as distribution and missing values are shown for each attribute, helping in better understanding of the data before applying the J48 algorithm.

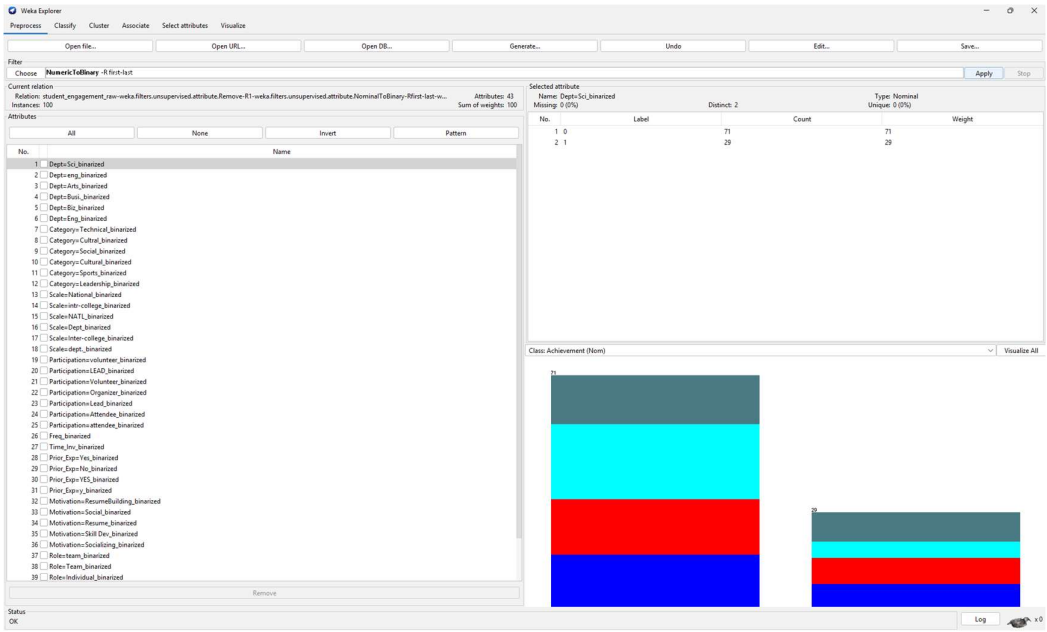

Figure 7.2: Conversion of Categorical Attributes using NominalToBinary Filter in WEKA

The dataset is displayed in WEKA's Preprocess tab after applying data transformation techniques. The **NominalToBinary filter** has been used to convert categorical attributes into binary format, resulting in multiple binarized attributes. The selected attribute shows binary values (0 and 1) with their distribution. The class attribute is set to **Achievement**, and the visualization panel displays the distribution of class labels. This transformation prepares the dataset for effective classification using the J48 algorithm.

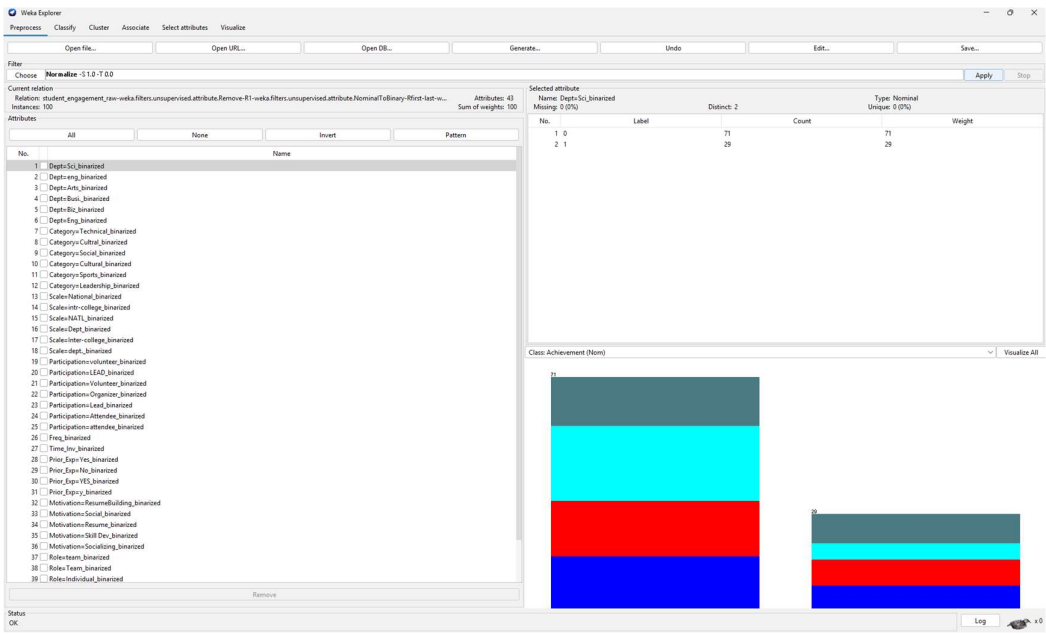

Figure 7.3: Data Normalization using Normalize Filter in WEKA

The dataset is displayed in WEKA's Preprocess tab after applying normalization. The **Normalize filter** is selected to scale all numeric attribute values within a standard range, improving consistency across the dataset. The attributes have already been converted into binary format, and no missing values are present. The class attribute is set to **Achievement**, and the visualization panel shows the distribution of class labels. This step ensures that the dataset is properly scaled and ready for accurate classification using the J48 algorithm.

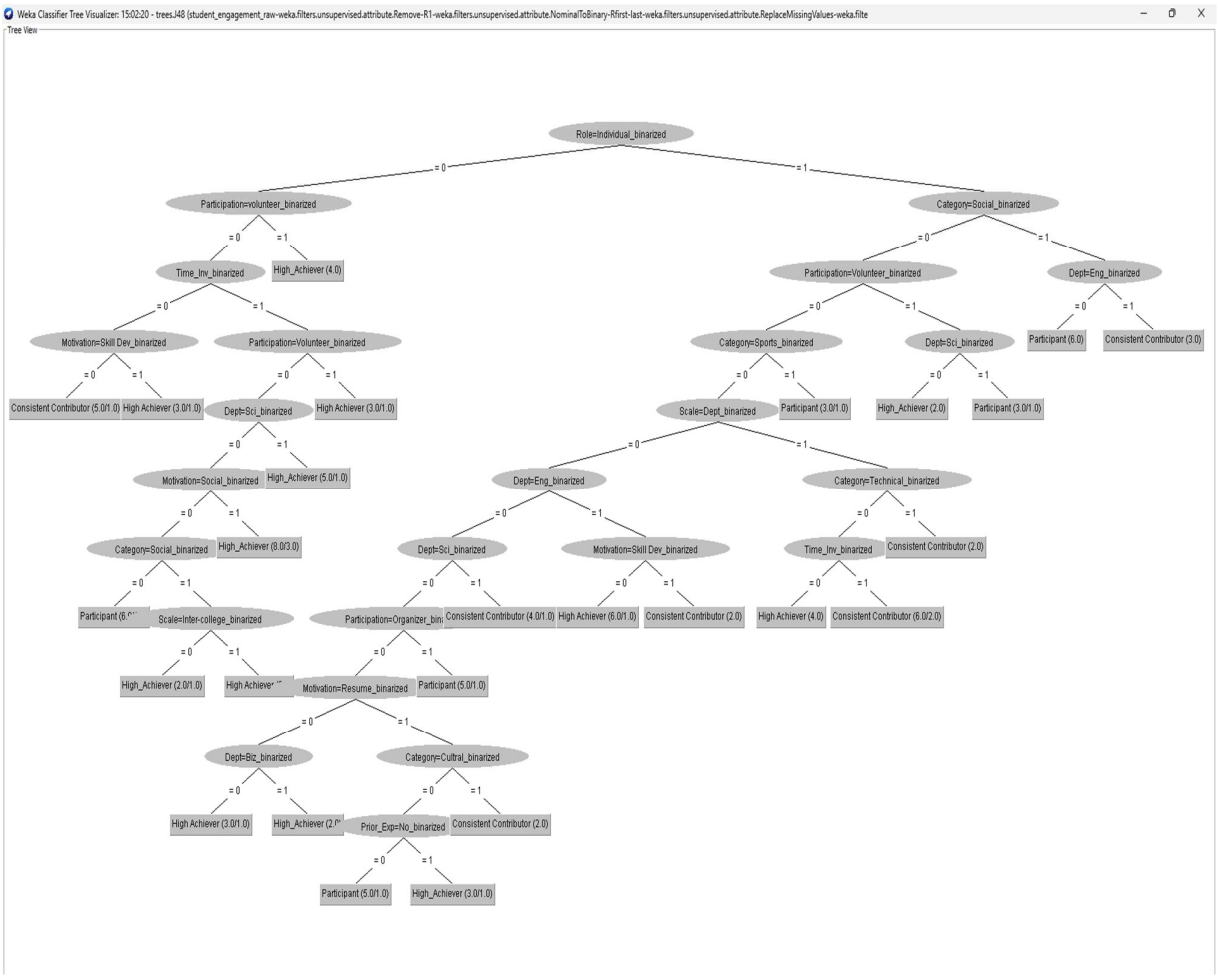

Figure 7.4: *Decision Tree Visualization using J48 Algorithm in WEKA*

This decision tree predicts student achievement using **Role_Individual**, followed by **Participation_Volunteer**, **Category_Social**, and **Dept_Eng** attributes.

If **Role_Individual = 0**, it checks Participation; if participation is high, it predicts **High Achiever**, otherwise it further evaluates time investment and motivation.

If **Role_Individual = 1**, it checks Category; if Category is Social, it evaluates department and predicts **Consistent Contributor** or **Participant**. In some branches, attributes like **Motivation**, **Scale**, and **Prior Experience** are used to refine the prediction between **High Achiever**, **Participant**, and **Consistent Contributor**.

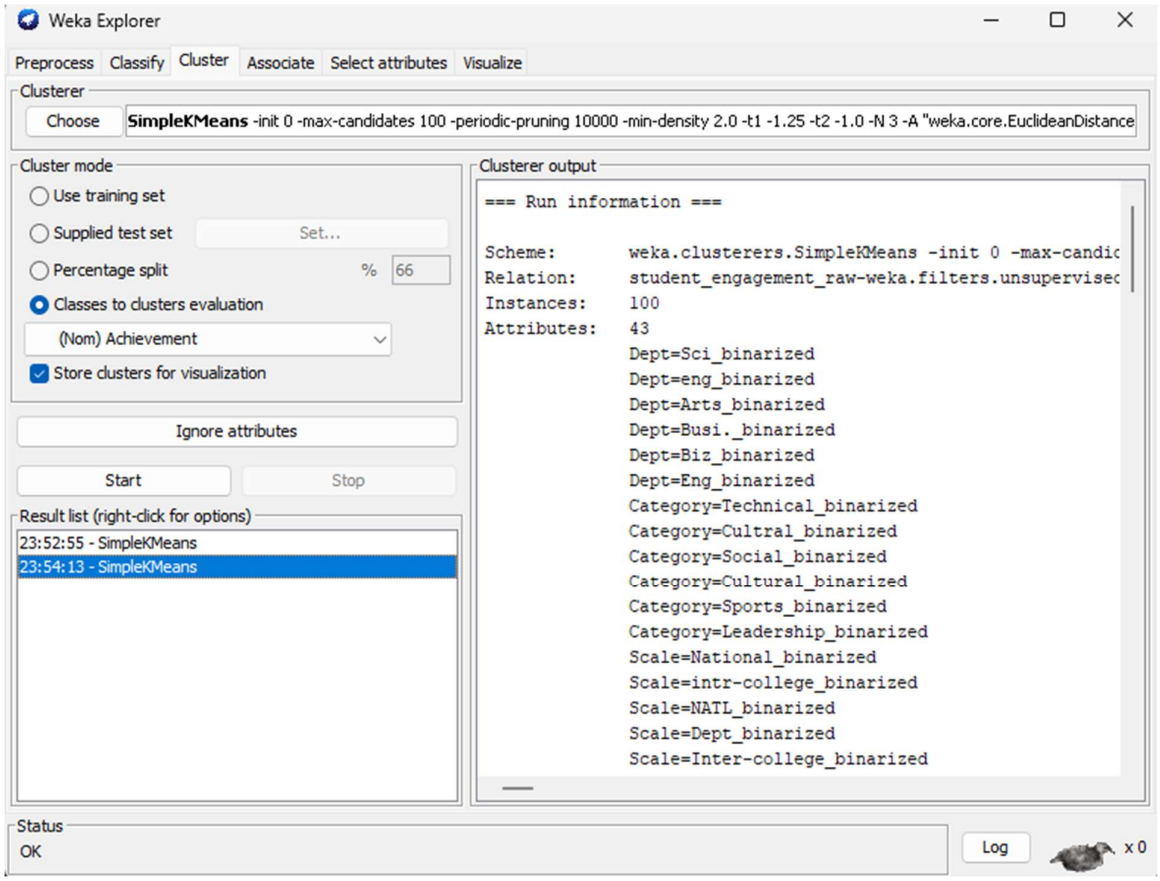

Figure 7.5: *K-Means Clustering using SimpleKMeans Algorithm in WEKA*

The dataset is processed in WEKA's Cluster tab using the SimpleKMeans algorithm to perform clustering. The "Classes to clusters evaluation" option is selected with Achievement as the reference attribute. The model groups the data into clusters based on similarities among attributes such as participation, motivation, and role. The clustering results, including cluster distribution and attribute details, are displayed in the output section, helping to identify patterns in student engagement.

8. CONCLUSION

In this project, student engagement data was successfully analyzed using the WEKA data mining tool. The dataset was preprocessed by handling missing values, converting categorical attributes into binary format, and normalizing the data to improve consistency. The J48 decision tree algorithm was applied to build a classification model for predicting student achievement levels.

The results demonstrate that attributes such as participation, motivation, role, and category significantly influence student performance. The decision tree provided clear and interpretable rules for classification. Although the model achieved moderate accuracy, it effectively highlights the relationship between engagement factors and academic outcomes.

Overall, this project shows how machine learning techniques can be used to analyze educational data and support better decision-making in academic environments.

The model achieved an accuracy of 27%, which can be improved by using larger datasets and advanced algorithms.

9. Future Enhancements

Real-time Data Integration – Integrate real-time student data from institutional systems for continuous performance monitoring and prediction.

Multi-class Prediction Enhancement – Improve classification by refining achievement categories and handling more detailed performance levels.

Improved Accuracy – Use advanced algorithms like Random Forest, Support Vector Machines, or Neural Networks to enhance prediction accuracy.

User Interface Development – Develop a web or mobile application to allow easy interaction and visualization of student performance data.

Personalized Recommendations – Provide suggestions to students based on their engagement patterns to improve performance.

Data Visualization & Insights – Use graphs and dashboards to visualize trends in student engagement and achievement.

Large Dataset Handling – Extend the system to handle larger datasets using big data technologies.

Automation – Automate data collection, preprocessing, and prediction processes for efficient analysis.

10. REFERENCES

WEKA Official Website – https://www.cs.waikato.ac.nz/ml/weka/

Data Mining: Concepts and Techniques – For understanding data mining concepts and algorithms

Machine Learning Basics – For basic machine learning theory and classification techniques

WEKA Documentation – For learning preprocessing, filters, and classification methods

GeeksforGeeks – For tutorials on data mining and machine learning concepts

TutorialsPoint – For reference materials on WEKA and classification algorithms

11. APPENDIX

```
=== Run information ===


Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 2

Relation:     student_engagement_raw-
weka.filters.unsupervised.attribute.Remove-R1-
weka.filters.unsupervised.attribute.NominalToBinary-
Rfirst-last-
weka.filters.unsupervised.attribute.ReplaceMissingValues-
weka.filters.unsupervised.attribute.NumericToBinary-
Rfirst-last-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-
weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0

Instances:    100

Attributes:   43

              Dept=Sci_binarized

              Dept=eng_binarized

              Dept=Arts_binarized

              Dept=Busi._binarized

              Dept=Biz_binarized

              Dept=Eng_binarized

              Category=Technical_binarized

              Category=Cultral_binarized

              Category=Social_binarized

              Category=Cultural_binarized
```

Category=Sports_binarized

Category=Leadership_binarized

Scale=National_binarized

Scale=intr-college_binarized

Scale=NATL_binarized

Scale=Dept_binarized

Scale=Inter-college_binarized

Scale=dept._binarized

Participation=volunteer_binarized

Participation=LEAD_binarized

Participation=Volunteer_binarized

Participation=Organizer_binarized

Participation=Lead_binarized

Participation=Attendee_binarized

Participation=attendee_binarized

Freq_binarized

Time_Inv_binarized

Prior_Exp=Yes_binarized

Prior_Exp=No_binarized

Prior_Exp=YES_binarized

Prior_Exp=y_binarized

Motivation=ResumeBuilding_binarized

Motivation=Social_binarized

Motivation=Resume_binarized

Motivation=Skill Dev_binarized

Motivation=Socializing_binarized

Role=team_binarized

Role=Team_binarized

Role=Individual_binarized

Mentor=No_binarized

Mentor=Yes_binarized

Mentor=YES_binarized

```
              Achievement

Test mode:    10-fold cross-validation


=== Classifier model (full training set) ===


J48 pruned tree

------------------


Role=Individual_binarized = 0

|   Participation=volunteer_binarized = 0

|   |   Time_Inv_binarized = 0

|   |   |   Motivation=Skill Dev_binarized = 0:
Consistent Contributor (5.0/1.0)

|   |   |   Motivation=Skill Dev_binarized = 1: High
Achiever (3.0/1.0)

|   |   Time_Inv_binarized = 1

|   |   |   Participation=Volunteer_binarized = 0

|   |   |   |   Dept=Sci_binarized = 0

|   |   |   |   |   Motivation=Social_binarized = 0

|   |   |   |   |   |   Category=Social_binarized = 0:
Participant (6.0/1.0)

|   |   |   |   |   |   Category=Social_binarized = 1

|   |   |   |   |   |   |   Scale=Inter-college_binarized
= 0: High_Achiever (2.0/1.0)

|   |   |   |   |   |   |   Scale=Inter-college_binarized
= 1: High Achiever (3.0)

|   |   |   |   |   Motivation=Social_binarized = 1:
High_Achiever (8.0/3.0)

|   |   |   |   Dept=Sci_binarized = 1: High_Achiever
(5.0/1.0)

|   |   |   Participation=Volunteer_binarized = 1: High
Achiever (3.0/1.0)

|   Participation=volunteer_binarized = 1: High_Achiever
(4.0)

Role=Individual_binarized = 1

|   Category=Social_binarized = 0
```

```
|   |   Participation=Volunteer_binarized = 0
|   |   |   Category=Sports_binarized = 0
|   |   |   |   Scale=Dept_binarized = 0
|   |   |   |   |   Dept=Eng_binarized = 0
|   |   |   |   |   |   Dept=Sci_binarized = 0
|   |   |   |   |   |   |
Participation=Organizer_binarized = 0
|   |   |   |   |   |   |   |
Motivation=Resume_binarized = 0
|   |   |   |   |   |   |   |   |   Dept=Biz_binarized =
0: High Achiever (3.0/1.0)
|   |   |   |   |   |   |   |   |   Dept=Biz_binarized =
1: High_Achiever (2.0)
|   |   |   |   |   |   |   |
Motivation=Resume_binarized = 1
|   |   |   |   |   |   |   |   |
Category=Cultral_binarized = 0
|   |   |   |   |   |   |   |   |   |
Prior_Exp=No_binarized = 0: Participant (5.0/1.0)
|   |   |   |   |   |   |   |   |   |
Prior_Exp=No_binarized = 1: High_Achiever (3.0/1.0)
|   |   |   |   |   |   |   |   |
Category=Cultral_binarized = 1: Consistent Contributor
(2.0)
|   |   |   |   |   |   |
Participation=Organizer_binarized = 1: Participant
(5.0/1.0)
|   |   |   |   |   |   Dept=Sci_binarized = 1:
Consistent Contributor (4.0/1.0)
|   |   |   |   |   Dept=Eng_binarized = 1
|   |   |   |   |   |   Motivation=Skill Dev_binarized =
0: High Achiever (6.0/1.0)
|   |   |   |   |   |   Motivation=Skill Dev_binarized =
1: Consistent Contributor (2.0)
|   |   |   |   Scale=Dept_binarized = 1
|   |   |   |   |   Category=Technical_binarized = 0
|   |   |   |   |   |   Time_Inv_binarized = 0: High
Achiever (4.0)
```

```
|   |   |   |   |   |   Time_Inv_binarized = 1:
Consistent Contributor (6.0/2.0)

|   |   |   |   |   Category=Technical_binarized = 1:
Consistent Contributor (2.0)

|   |   |   Category=Sports_binarized = 1: Participant
(3.0/1.0)

|   |   Participation=Volunteer_binarized = 1

|   |   |   Dept=Sci_binarized = 0: High_Achiever (2.0)

|   |   |   Dept=Sci_binarized = 1: Participant (3.0/1.0)

|   Category=Social_binarized = 1

|   |   Dept=Eng_binarized = 0: Participant (6.0)

|   |   Dept=Eng_binarized = 1: Consistent Contributor
(3.0)


Number of Leaves  :   26


Size of the tree :    51



Time taken to build model: 0.01 seconds


=== Stratified cross-validation ===

=== Summary ===


Correctly Classified Instances          27
27      %

Incorrectly Classified Instances        73
73      %

Kappa statistic                          0.0276

Mean absolute error                      0.3814

Root mean squared error                  0.5569

Relative absolute error                101.7636 %

Root relative squared error            128.5655 %

Total Number of Instances              100
```

```
=== Detailed Accuracy By Class ===


                 TP Rate  FP Rate  Precision  Recall   F-
Measure  MCC      ROC Area  PRC Area  Class

                 0.304    0.273    0.250      0.304
0.275      0.030    0.514     0.240     High_Achiever

                 0.280    0.293    0.241      0.280
0.259      -0.013   0.481     0.242     High Achiever

                 0.214    0.236    0.261      0.214
0.235      -0.023   0.425     0.269     Participant

                 0.292    0.171    0.350      0.292
0.318      0.129    0.505     0.255     Consistent
Contributor

Weighted Avg.    0.270    0.243    0.275      0.270
0.270      0.028    0.479     0.252


=== Confusion Matrix ===


  a  b  c  d   <-- classified as

  7  6  5  5 |  a = High_Achiever

  8  7  6  4 |  b = High Achiever

  8 10  6  4 |  c = Participant

  5  6  6  7 |  d = Consistent Contributor


=== Source code ===


// Generated with Weka 3.8.6

//

// This code is public domain and comes with no warranty.

//

// Timestamp: Sun Mar 22 20:46:57 IST 2026


package weka.classifiers;


import weka.core.Attribute;
```

```
import weka.core.Capabilities;
import weka.core.Capabilities.Capability;
import weka.core.Instance;
import weka.core.Instances;
import weka.core.RevisionUtils;
import weka.classifiers.Classifier;
import weka.classifiers.AbstractClassifier;

public class WekaWrapper
  extends AbstractClassifier {

  /**
   * Returns only the toString() method.
   *
   * @return a string describing the classifier
   */
  public String globalInfo() {
    return toString();
  }

  /**
   * Returns the capabilities of this classifier.
   *
   * @return the capabilities
   */
  public Capabilities getCapabilities() {
    weka.core.Capabilities result = new
weka.core.Capabilities(this);


result.enable(weka.core.Capabilities.Capability.NOMINAL_A
TTRIBUTES);
```

```
result.enable(weka.core.Capabilities.Capability.NUMERIC_A
TTRIBUTES);

result.enable(weka.core.Capabilities.Capability.DATE_ATTR
IBUTES);

result.enable(weka.core.Capabilities.Capability.MISSING_V
ALUES);

result.enable(weka.core.Capabilities.Capability.NOMINAL_C
LASS);

result.enable(weka.core.Capabilities.Capability.MISSING_C
LASS_VALUES);



    result.setMinimumNumberInstances(0);


    return result;
  }


  /**
   * only checks the data against its capabilities.
   *
   * @param i the training data
   */
  public void buildClassifier(Instances i) throws
Exception {
    // can classifier handle the data?
    getCapabilities().testWithFail(i);
  }


  /**
   * Classifies the given instance.
   *
   * @param i the instance to classify
```

```
   * @return the classification result
   */
  public double classifyInstance(Instance i) throws
Exception {
    Object[] s = new Object[i.numAttributes()];

    for (int j = 0; j < s.length; j++) {
      if (!i.isMissing(j)) {
        if (i.attribute(j).isNominal())
          s[j] = new String(i.stringValue(j));
        else if (i.attribute(j).isNumeric())
          s[j] = new Double(i.value(j));
      }
    }

    // set class value to missing
    s[i.classIndex()] = null;

    return WekaClassifier.classify(s);
  }

  /**
   * Returns the revision string.
   *
   * @return        the revision
   */
  public String getRevision() {
    return RevisionUtils.extract("1.0");
  }

  /**
   * Returns only the classnames and what classifier it
is based on.
```

```
   *
   * @return a short description
   */
  public String toString() {
    return "Auto-generated classifier wrapper, based on
weka.classifiers.trees.J48 (generated with Weka
3.8.6).\n" + this.getClass().getName() +
"/WekaClassifier";
  }

  /**
   * Runs the classfier from commandline.
   *
   * @param args the commandline arguments
   */
  public static void main(String args[]) {
    runClassifier(new WekaWrapper(), args);
  }
}

class WekaClassifier {

  public static double classify(Object[] i)
    throws Exception {

    double p = Double.NaN;
    p = WekaClassifier.N48e49e0c0(i);
    return p;
  }
  static double N48e49e0c0(Object []i) {
    double p = Double.NaN;
    if (i[38] == null) {
      p = 0;
    } else if (i[38].equals("0")) {
```

```
  p = WekaClassifier.N29ecc8f1(i);
  } else if (i[38].equals("1")) {
  p = WekaClassifier.N291293fa9(i);
  }
  return p;
}
static double N29ecc8f1(Object []i) {
  double p = Double.NaN;
  if (i[18] == null) {
    p = 0;
  } else if (i[18].equals("0")) {
  p = WekaClassifier.N685c8db02(i);
  } else if (i[18].equals("1")) {
    p = 0;
  }
  return p;
}
static double N685c8db02(Object []i) {
  double p = Double.NaN;
  if (i[26] == null) {
    p = 3;
  } else if (i[26].equals("0")) {
  p = WekaClassifier.N35176ef43(i);
  } else if (i[26].equals("1")) {
  p = WekaClassifier.N596f92a44(i);
  }
  return p;
}
static double N35176ef43(Object []i) {
  double p = Double.NaN;
  if (i[34] == null) {
    p = 3;
```

```
  } else if (i[34].equals("0")) {
    p = 3;
  } else if (i[34].equals("1")) {
    p = 1;
  }
  return p;
}
static double N596f92a44(Object []i) {
  double p = Double.NaN;
  if (i[20] == null) {
    p = 0;
  } else if (i[20].equals("0")) {
  p = WekaClassifier.N2de1c7b25(i);
  } else if (i[20].equals("1")) {
    p = 1;
  }
  return p;
}
static double N2de1c7b25(Object []i) {
  double p = Double.NaN;
  if (i[0] == null) {
    p = 0;
  } else if (i[0].equals("0")) {
  p = WekaClassifier.N215ceeb36(i);
  } else if (i[0].equals("1")) {
    p = 0;
  }
  return p;
}
static double N215ceeb36(Object []i) {
  double p = Double.NaN;
  if (i[32] == null) {
```

```
    p = 2;
  } else if (i[32].equals("0")) {
  p = WekaClassifier.N2f3244a27(i);
  } else if (i[32].equals("1")) {
    p = 0;
  }
  return p;
}
static double N2f3244a27(Object []i) {
  double p = Double.NaN;
  if (i[8] == null) {
    p = 2;
  } else if (i[8].equals("0")) {
    p = 2;
  } else if (i[8].equals("1")) {
  p = WekaClassifier.Na6d09528(i);
  }
  return p;
}
static double Na6d09528(Object []i) {
  double p = Double.NaN;
  if (i[16] == null) {
    p = 0;
  } else if (i[16].equals("0")) {
    p = 0;
  } else if (i[16].equals("1")) {
    p = 1;
  }
  return p;
}
static double N291293fa9(Object []i) {
  double p = Double.NaN;
```

```
  if (i[8] == null) {
    p = 1;
  } else if (i[8].equals("0")) {
  p = WekaClassifier.N6ffdc91d10(i);
  } else if (i[8].equals("1")) {
  p = WekaClassifier.N2fa04d324(i);
  }
  return p;
}
static double N6ffdc91d10(Object []i) {
  double p = Double.NaN;
  if (i[20] == null) {
    p = 1;
  } else if (i[20].equals("0")) {
  p = WekaClassifier.N5257971411(i);
  } else if (i[20].equals("1")) {
  p = WekaClassifier.N728b647c23(i);
  }
  return p;
}
static double N5257971411(Object []i) {
  double p = Double.NaN;
  if (i[10] == null) {
    p = 1;
  } else if (i[10].equals("0")) {
  p = WekaClassifier.N3092239512(i);
  } else if (i[10].equals("1")) {
    p = 2;
  }
  return p;
}
static double N3092239512(Object []i) {
```

```
  double p = Double.NaN;
  if (i[15] == null) {
    p = 2;
  } else if (i[15].equals("0")) {
  p = WekaClassifier.Nda2fb3e13(i);
  } else if (i[15].equals("1")) {
  p = WekaClassifier.N28c7976121(i);
  }
  return p;
}
static double Nda2fb3e13(Object []i) {
  double p = Double.NaN;
  if (i[5] == null) {
    p = 2;
  } else if (i[5].equals("0")) {
  p = WekaClassifier.Nc166eb614(i);
  } else if (i[5].equals("1")) {
  p = WekaClassifier.N3f39f70120(i);
  }
  return p;
}
static double Nc166eb614(Object []i) {
  double p = Double.NaN;
  if (i[0] == null) {
    p = 2;
  } else if (i[0].equals("0")) {
  p = WekaClassifier.N6217b71715(i);
  } else if (i[0].equals("1")) {
    p = 3;
  }
  return p;
}
```

```
static double N6217b71715(Object []i) {
  double p = Double.NaN;
  if (i[21] == null) {
    p = 0;
  } else if (i[21].equals("0")) {
  p = WekaClassifier.N43b8ae4b16(i);
  } else if (i[21].equals("1")) {
    p = 2;
  }
  return p;
}
static double N43b8ae4b16(Object []i) {
  double p = Double.NaN;
  if (i[33] == null) {
    p = 0;
  } else if (i[33].equals("0")) {
  p = WekaClassifier.N59d9c60917(i);
  } else if (i[33].equals("1")) {
  p = WekaClassifier.N6605605a18(i);
  }
  return p;
}
static double N59d9c60917(Object []i) {
  double p = Double.NaN;
  if (i[4] == null) {
    p = 1;
  } else if (i[4].equals("0")) {
    p = 1;
  } else if (i[4].equals("1")) {
    p = 0;
  }
  return p;
```

```
}
static double N6605605a18(Object []i) {
  double p = Double.NaN;
  if (i[7] == null) {
    p = 2;
  } else if (i[7].equals("0")) {
  p = WekaClassifier.N39f5cc8219(i);
  } else if (i[7].equals("1")) {
    p = 3;
  }
  return p;
}
static double N39f5cc8219(Object []i) {
  double p = Double.NaN;
  if (i[28] == null) {
    p = 2;
  } else if (i[28].equals("0")) {
    p = 2;
  } else if (i[28].equals("1")) {
    p = 0;
  }
  return p;
}
static double N3f39f70120(Object []i) {
  double p = Double.NaN;
  if (i[34] == null) {
    p = 1;
  } else if (i[34].equals("0")) {
    p = 1;
  } else if (i[34].equals("1")) {
    p = 3;
  }
```

```
  return p;
}
static double N28c7976121(Object []i) {
  double p = Double.NaN;
  if (i[6] == null) {
    p = 1;
  } else if (i[6].equals("0")) {
  p = WekaClassifier.N658d259322(i);
  } else if (i[6].equals("1")) {
    p = 3;
  }
  return p;
}
static double N658d259322(Object []i) {
  double p = Double.NaN;
  if (i[26] == null) {
    p = 1;
  } else if (i[26].equals("0")) {
    p = 1;
  } else if (i[26].equals("1")) {
    p = 3;
  }
  return p;
}
static double N728b647c23(Object []i) {
  double p = Double.NaN;
  if (i[0] == null) {
    p = 0;
  } else if (i[0].equals("0")) {
    p = 0;
  } else if (i[0].equals("1")) {
    p = 2;
```

```
    }
    return p;
  }
  static double N2fa04d324(Object []i) {
    double p = Double.NaN;
    if (i[5] == null) {
      p = 2;
    } else if (i[5].equals("0")) {
      p = 2;
    } else if (i[5].equals("1")) {
      p = 3;
    }
    return p;
  }
}
```

By:

24B11AI019

24B11AI044

24B11AI135

24B11AI146